Framewise displacement (FD) vs each electrogastrogram (EGG) channel

1 Goal

The main pipeline uses a single dominant electrogastrogram (EGG) channel (the one with the highest in-band power) selected during preprocessing. Here we look at all four channels to check two things:

1. How framewise displacement (FD, Power et al. 2012) relates to each individual EGG channel.
2. Whether the choice of dominant channel affects the FD vs EGG coupling result.

The implementation lives in `fwd_vs_each_egg_channel.py`; outputs are in `outputs/`.

2 Pipeline

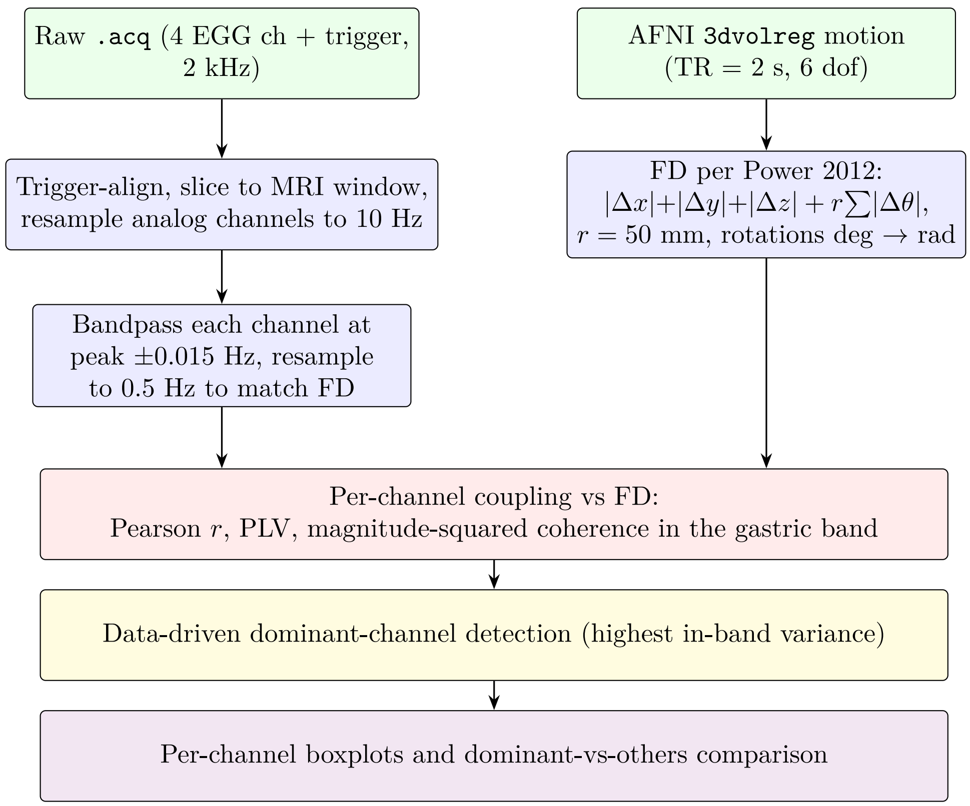

Figure 1: Per-run pipeline. Green = data, blue = computational steps; the wide blocks below are the per-channel coupling step (red), the data-driven dominant-channel detection (yellow), and the group-level visualisation (purple). The two top columns merge into the coupling block: each EGG channel contributes one column of statistics per run.

3 Data

- The same 84 resting-state runs from 43 healthy adults used in the PVAF analysis. Re-analysed from the dataset shared by Levakov, Ganor and Avidan (2023).
- Raw 4-channel cutaneous EGG (`physio/{sub}/egg/{sub}_rest{run}.acq`), recorded at 2 kHz alongside a magnetic resonance imaging (MRI) trigger digital channel. Some runs have only 3 active EGG electrodes (metadata column `num_channles`), giving 327 (run, channel) pairs in total.
- AFNI `3dvolreg` motion (6 degrees of freedom (dof), repetition time TR = 2 s) from the same project files used in the percent variance accounted for (PVAF) analysis.

4 Method

4.1 Framewise displacement (Power et al. 2012, eq. 2)

Framewise displacement (FD) is a single scalar time course that summarises total head motion. It is computed step by step as follows for each run:

1. Read the six AFNI 1D motion columns (three translations and three rotations) at TR = 2 s.
2. Convert the three rotation columns from degrees (the AFNI `3dvolreg` output) to radians and multiply by the head-sphere radius $r = 50\,\text{mm}$, so that all six columns are in millimetres.
3. Take the first frame-to-frame difference of each column.
4. Sum the absolute values of the six differences at each time point.

In equations,

$$\mathrm{FD}(t) \;=\; \underbrace{|\Delta x(t)| + |\Delta y(t)| + |\Delta z(t)|}_{\text{translation, mm}} \;+\; r\cdot \underbrace{\big(|\Delta\theta_{\text{roll}}(t)| + |\Delta\theta_{\text{pitch}}(t)| + |\Delta\theta_{\text{yaw}}(t)|\big)}_{\text{rotation, rad}}. \tag{1}$$

where:

- t is the discrete time index (one sample per repetition time);
- $\Delta x(t) = x(t) - x(t-1)$ is the frame-to-frame change in the x translation (and similarly for y, z), in millimetres;
- $\Delta\theta_\bullet(t) = \theta_\bullet(t) - \theta_\bullet(t-1)$ is the corresponding frame-to-frame change in rotation about the roll, pitch, or yaw axis, in radians;
- $|\cdot|$ denotes the absolute value;
- $r = 50\,\text{mm}$ is the sphere radius assumed for the head, used to convert angular displacement into an equivalent arc length, following Power et al. (2012).

The resulting FD is a single non-negative time course in millimetres at TR = 2 s.

4.2 Per-channel preprocessing

For each (subject, run) the EGG side of the analysis is built in four steps:

1. Read the raw `.acq` file with `bioread` at its native 2 kHz sampling rate. The file contains the four EGG analog channels and a digital MRI-trigger channel.
2. Locate the MRI trigger onset (the first sample where the trigger channel goes high) using the same procedure as `preprocess_gastric.py`, then slice every analog channel to the trigger-locked MRI window, whose length in samples is given by the metadata column `mri_length`.
3. Resample each analog channel from 2 kHz down to 10 Hz using `scipy.signal.resample` (Fourier resampling).
4. Bandpass-filter each 10 Hz channel at the subject-specific gastric peak ±0.015 Hz using the same zero-phase FIR filter (MNE `filter_data`) as the rest of the pipeline. The peak frequency is taken from `max_freq*.npy`, the file produced by `preprocess_gastric.py`.
5. Resample the bandpassed channel from 10 Hz down to $f_s = 0.5\,\text{Hz}$ so it shares a time grid with FD; truncate both series to the shorter of the two if their lengths differ.

4.3 Coupling metrics per channel

For each channel c and each run, three coupling statistics are recorded between FD and the bandpassed channel c_{bp}:

- **Pearson correlation coefficient r.** Defined for two time series $a(t)$, $b(t)$ of length T as

$$r(a,b) \;=\; \frac{\sum_{t=1}^{T}\big(a(t)-\bar{a}\big)\big(b(t)-\bar{b}\big)}{\sqrt{\sum_{t=1}^{T}(a(t)-\bar{a})^2}\;\sqrt{\sum_{t=1}^{T}(b(t)-\bar{b})^2}} \;\in\; [-1,\,+1], \tag{2}$$

  where $\bar{a}$ and $\bar{b}$ are the time-means of a and b and r is signed. Here $a = \text{FD}$ and $b = c_{\text{bp}}$. The value is computed with `scipy.stats.pearsonr`.

- **Phase-locking value (PLV).** A measure of how stable the phase difference between two signals is across time:

$$\text{PLV} \;=\; \left|\frac{1}{T}\sum_{t=1}^{T} e^{\,j\left(\phi_{\text{FD}}(t)-\phi_c(t)\right)}\right|, \tag{3}$$

  where:

  - T is the number of time samples in the run;
  - $j = \sqrt{-1}$ is the imaginary unit;
  - $\phi_{\text{FD}}(t)$ is the instantaneous phase of FD at time t, obtained from `np.angle` applied to `scipy.signal.hilbert`;
  - $\phi_c(t)$ is the same quantity computed from the bandpassed EGG channel c;
  - $|\cdot|$ denotes the modulus of the complex average.

  PLV is bounded between 0 and 1: PLV $= 1$ corresponds to a constant phase difference across the run, and PLV $= 0$ to phase differences that are uniformly distributed on the unit circle.

- **Magnitude-squared coherence in the gastric band.** Coherence is the frequency-domain analogue of correlation. For frequency f it is defined as

$$C_{ab}(f) \;=\; \frac{|S_{ab}(f)|^2}{S_{aa}(f)\,S_{bb}(f)} \;\in\; [0,\,1], \tag{4}$$

where $S_{aa}(f)$ and $S_{bb}(f)$ are the Welch power spectra of a and b and $S_{ab}(f)$ is their Welch cross-spectrum (all estimated by `scipy.signal.coherence`). The reported value is the mean of $C_{ab}(f)$ over the frequency bins that fall inside the gastric band $[f_{\text{peak}} - 0.015,\ f_{\text{peak}} + 0.015]$ Hz.

4.4 Dominant-channel detection

The metadata column `dominant_channel` is set to the literal string `"auto"` for most runs and is not back-filled with the channel index that the OHBM pipeline actually selected. To recover an explicit choice we re-detect the dominant channel from the data itself, using the rule the OHBM preprocessing follows in `preprocess_gastric.py` but applied directly to the bandpassed signal:

1. For each of the four EGG channels, take the bandpassed signal at 10 Hz from the previous step and compute its time-domain variance $\sigma_c^2 = \mathrm{Var}(c_{\text{bp,10Hz}})$.
2. Set the dominant channel index to $c^* = \arg\max_c \sigma_c^2$.

Because the signal is already bandpassed at the subject's gastric peak ± 0.015 Hz, the variance is exactly the in-band power (Parseval's theorem), which is what `preprocess_gastric.py` picks via Welch peak power.

The procedure gives exactly one dominant channel per run. Across the 84 runs this yields 84 rows flagged `is_dominant = True` in `outputs/fwd_per_channel_results.csv` and 243 rows with `False` (some runs have only three active EGG electrodes, so the total row count is $84 + 243 = 327$ rather than $4 \times 84 = 336$).

5 Results

5.1 FD sanity check

For healthy adults at TR = 2 s the literature reports mean FD in the range 0.05 to 0.30 mm and max FD typically below 0.5 mm (Power et al. 2012; Satterthwaite et al. 2013). On this dataset:

FD statistic	median across 84 runs	IQR
mean FD (mm)	**0.104**	0.085–0.131
max FD (mm)	**0.458**	0.313–0.644

Both fall well inside the published range, which supports that the FD computation is correct.

5.2 Example time-course overlay

The figure below stacks FD on top and the four bandpassed EGG channels underneath for one example run (`LA`, run 1, gastric peak $\approx$ 0.0507 Hz). All four channels carry the same $\sim$0.05 Hz rhythm with very similar morphology, which is what one would expect for nearby cutaneous electrodes recording the same gastric slow wave. FD on top does not visually track those oscillations cycle for cycle, which is consistent with the small statistical effect found in the PVAF analysis.

Two further example overlays are in `outputs/fwd_overlay_LA_run2.png` and `outputs/fwd_overlay_BS_run1.png`.

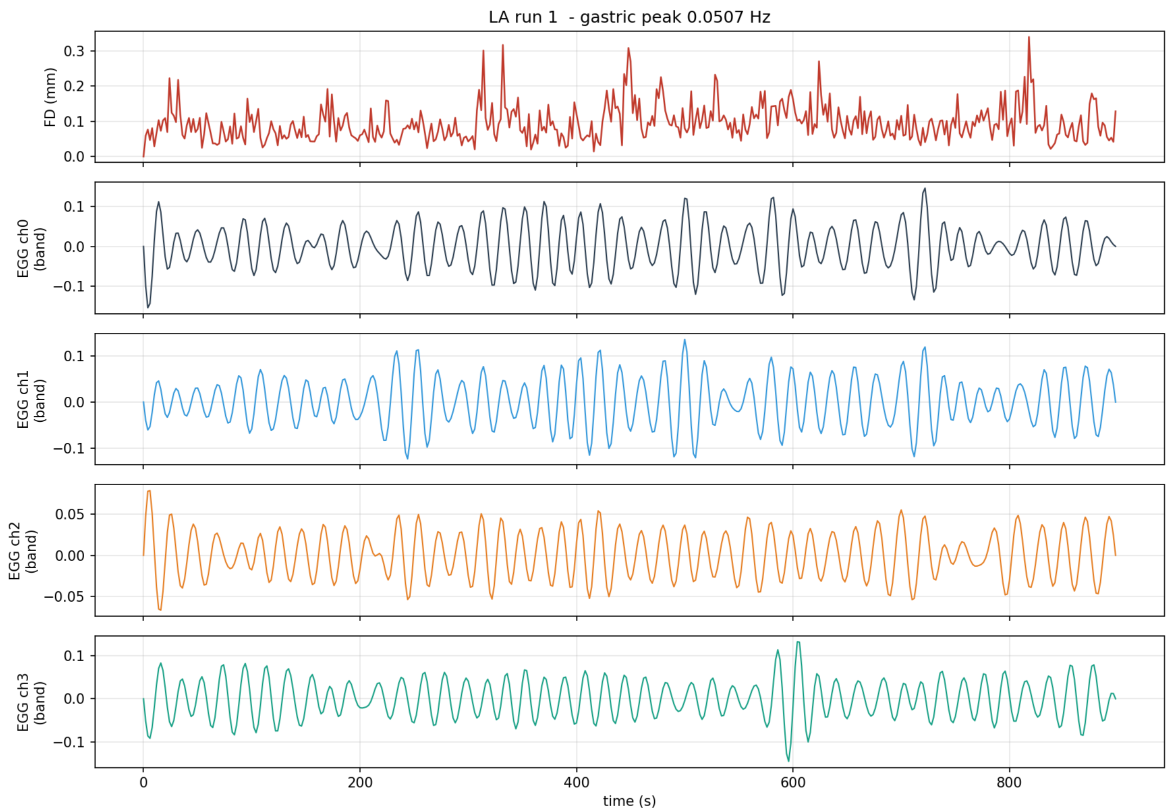

Figure 2: **Illustrative example from a single run** (`LA` run 1, $n = 1$): FD (red, top) and the four bandpassed EGG channels (below). Group-level summaries across all 84 runs are in figures 3 and 4.

5.3 Channel-by-channel coupling

Median coupling between FD and each individual EGG channel, across all available runs:

Channel	n	PLV (median)	Coherence (median)	Pearson r (median)
0	84	0.025	0.083	-0.008
1	84	0.024	0.093	-0.001
2	84	0.029	0.084	0.007
3	75	0.022	0.079	-0.008

The four channels look broadly interchangeable: PLV medians lie in a narrow band (0.022–0.029), coherence medians in 0.079–0.093, and Pearson r medians sit near zero. The full distributions are below.

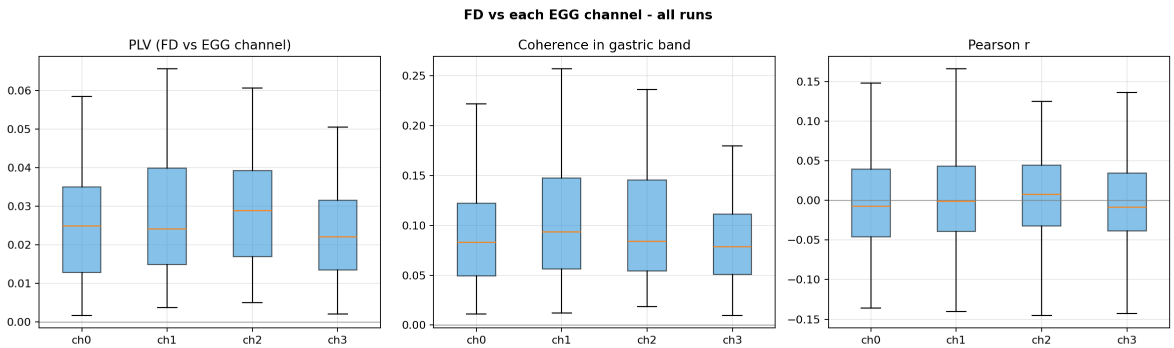

Figure 3: FD vs each EGG channel: per-channel boxplots of PLV (left), coherence in the gastric band (middle), and Pearson r (right). $n = 84$ **runs** per box (channels 0–2) and $n = 75$ **runs** for channel 3 (some runs only have 3 active EGG electrodes).

5.4 Does the dominant-channel choice matter?

If the OHBM pipeline's dominant-channel pick captured a special FD signal, the "dominant" boxes in the figure below would sit visibly above the "other electrodes" boxes. They do not.

	n	PLV (median)	Coherence (median)	Pearson r (median)
dominant electrode	84	0.026	0.080	−0.005
other electrodes	243	0.025	0.087	−0.001

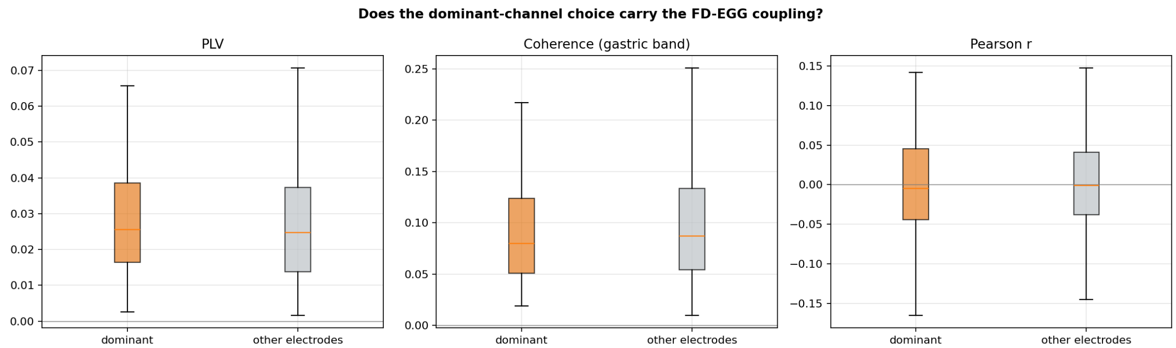

Figure 4: Dominant electrode (orange) vs the other electrodes (grey) for PLV (left), coherence in band (middle), and Pearson r (right). $n = 84$ **dominant rows,** $n = 243$ **other-electrode rows** across all 84 runs.

A separate way to ask the same question: in how many runs does the data-detected dominant channel coincide with the channel that has the strongest FD coupling?

- dominant = argmax over channels of FD-EGG PLV: 23 / 84 runs (27.4%).
- dominant = argmax over channels of FD-EGG coherence: 22 / 84 runs (26.2%).

With 4 channels the chance rate is 25%, so the dominant pick does not appear to track the FD-coupled channel. That may suggest the channel that carries the most EGG power and the channel that couples best with FD are roughly independent questions; either way, the choice of dominant channel does not appear to bias the FD-EGG coupling result up or down in a systematic way.

6 Summary

The mean and max FD on this dataset fall well within the range Power et al. (2012) and Satterthwaite et al. (2013) report for healthy adults at TR = 2 s. This observation may suggest that the FD implementation behaves as expected for typical resting-state data.

The four cutaneous EGG channels appear to record the same ∼0.05 Hz gastric slow wave: the bandpassed time courses overlay closely (figure 2) and the coupling-with-FD distributions look broadly interchangeable across channels.

The data-driven dominant channel (the one with the highest in-band EGG variance) does not, on this dataset, coincide with the channel that shows the strongest FD coupling more often than chance (about 26 to 27% for 4 channels, compared with the 25% chance rate). The "dominant" and "other electrodes" boxplots look similar across all three coupling metrics. Together, these observations may suggest that the OHBM pipeline's dominant-channel pick does not bias the FD vs EGG coupling result up or down in any systematic way for these runs, and that the four electrodes can be treated as roughly interchangeable for this specific question.

References

- Power JD *et al.* (2012). Spurious but systematic correlations in functional connectivity MRI networks arise from subject motion. *NeuroImage* 59(3):2142–2154.
- Satterthwaite TD *et al.* (2013). An improved framework for confound regression and filtering for control of motion artifact in the preprocessing of resting-state functional connectivity data. *NeuroImage* 64:240–256.
- Choe AS *et al.* (2021). Phase-locking of resting-state brain networks with the gastric basal electrical rhythm. *PLoS ONE* 16(1):e0244756.
- Levakov G, Ganor S, Avidan G (2023). Reliability and validity of brain–gastric phase synchronization. *Human Brain Mapping* 44:4956–4966.